

EEE 591 : Python for Rapid Engineering Solutions

Project 1 Report

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PROBLEM 1: DATA ANALYSIS

1.1 Dataset Overview

The heart disease dataset (heart1.csv) contains 270 patient records with 13 medical features and 1 target variable (a1p2). The target variable indicates absence (1) or presence (2) of heart disease. The dataset contains no missing values across any of the 14 columns, making it clean and ready for analysis without imputation.

Property	Value
File	heart1.csv
Total Records	270
Total Features	13 (all used, none dropped, no PCA)
Target Variable	a1p2 (1 = No Heart Disease, 2 = Heart Disease)
Missing Values	0

1.2 Dataset Variables

Num	Name	Description
0	age	Patient age
1	sex	Sex (1=male, 0=female)
2	cpt	Chest pain type (4 values)
3	rbp	Resting blood pressure
4	sc	Serum cholesterol in mg/dl
5	fbs	Fasting blood sugar > 120 mg/dl
6	rer	Resting electrocardiographic results (0,1,2)
7	mhr	Maximum heart rate achieved
8	eia	Exercise induced angina
9	opst	ST depression induced by exercise relative to rest
10	dests	Slope of the peak exercise ST segment
11	nmvcf	Number of major vessels (0-3) colored by fluoroscopy
12	thal	Thallium stress test (3=normal, 6=fixed, 7=reversible)
13	a1p2	TARGET: 1=no heart disease, 2=heart disease

1.3 Descriptive Statistics

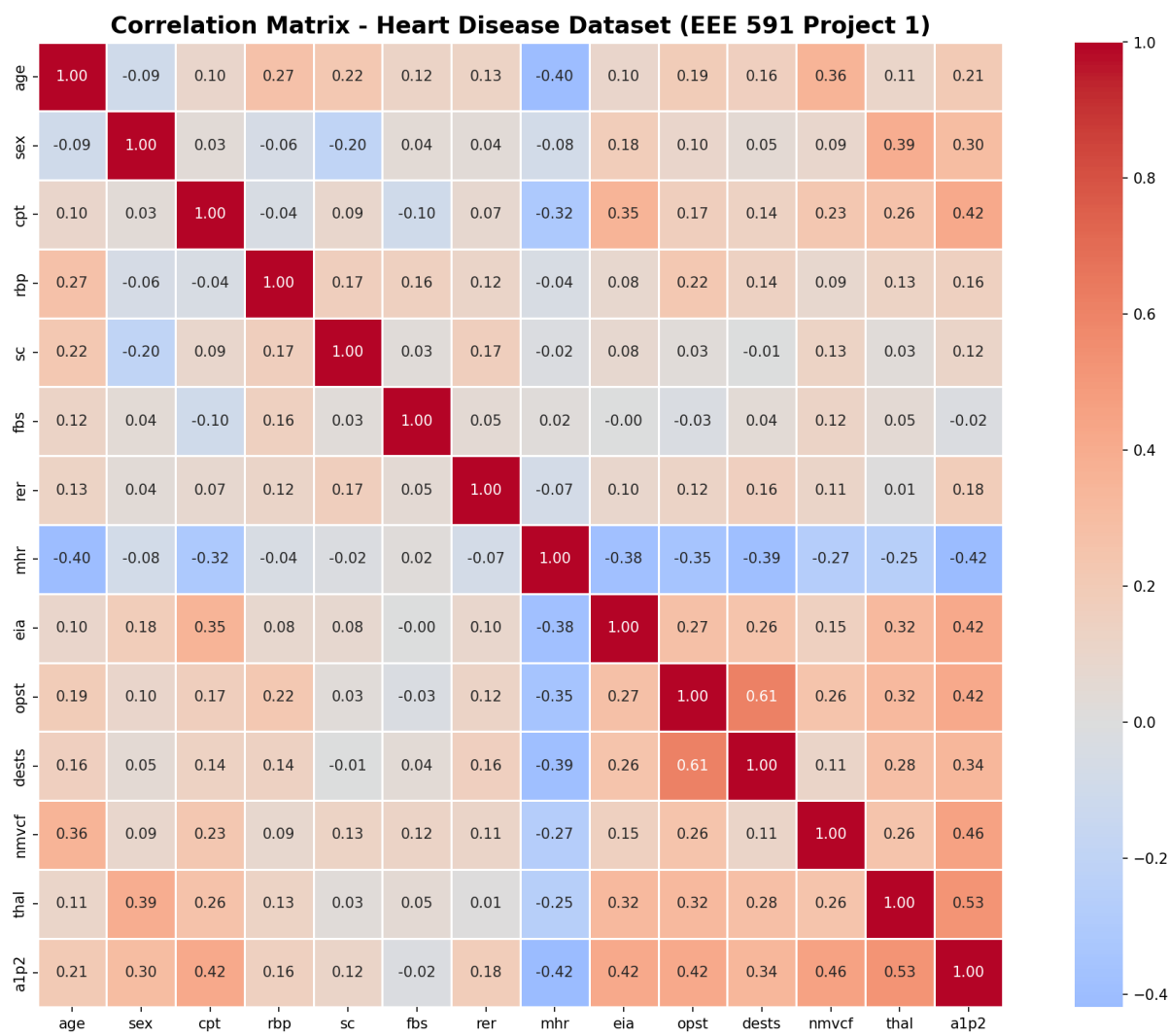
Stat	age	sex	cpt	rbp	sc	fbs	rer	mhr	eia	opst	dests	nmvcf	thal
count	270	270	270	270	270	270	270	270	270	270	270	270	270
mean	54.43	0.68	3.17	131.34	249.66	0.15	1.02	149.68	0.33	1.05	1.59	0.67	4.70
std	9.11	0.47	0.95	17.86	51.69	0.36	1.00	23.17	0.47	1.15	0.61	0.94	1.94
min	29.00	0.00	1.00	94.00	126.00	0.00	0.00	71.00	0.00	0.00	1.00	0.00	3.00
25%	48.00	0.00	3.00	120.00	213.00	0.00	0.00	133.00	0.00	0.00	1.00	0.00	3.00
50%	55.00	1.00	3.00	130.00	245.00	0.00	2.00	153.50	0.00	0.80	2.00	0.00	3.00
75%	61.00	1.00	4.00	140.00	280.00	0.00	2.00	166.00	1.00	1.60	2.00	1.00	7.00
max	77.00	1.00	4.00	200.00	564.00	1.00	2.00	202.00	1.00	6.20	3.00	3.00	7.00

1.4 Correlation of Each Feature with Target (a1p2)

Sorted by absolute correlation value descending. Statistical significance tested using point-biserial correlation with $\alpha = 0.05$.

Feature	Pearson r	r	p-value	Significant?
thal	+0.5250	0.5250	1.57e-20	Yes (p<0.05)
nmvcf	+0.4553	0.4553	3.17e-15	Yes (p<0.05)
eia	+0.4193	0.4193	6.38e-13	Yes (p<0.05)
mhr	-0.4185	0.4185	7.12e-13	Yes (p<0.05)
opst	+0.4180	0.4180	7.68e-13	Yes (p<0.05)
cpt	+0.4174	0.4174	8.26e-13	Yes (p<0.05)
dests	+0.3376	0.3376	1.27e-08	Yes (p<0.05)
sex	+0.2977	0.2977	6.27e-07	Yes (p<0.05)
age	+0.2123	0.2123	4.43e-04	Yes (p<0.05)
rer	+0.1821	0.1821	2.67e-03	Yes (p<0.05)
rbp	+0.1554	0.1554	1.06e-02	Yes (p<0.05)
sc	+0.1180	0.1180	5.27e-02	No
fbs	-0.0163	0.0163	7.90e-01	No

1.5 Full Correlation Matrix - Heatmap



1.6 Top 5 Feature Pairs Most Correlated With Each Other

Excluding self-correlation and target variable (alp2).

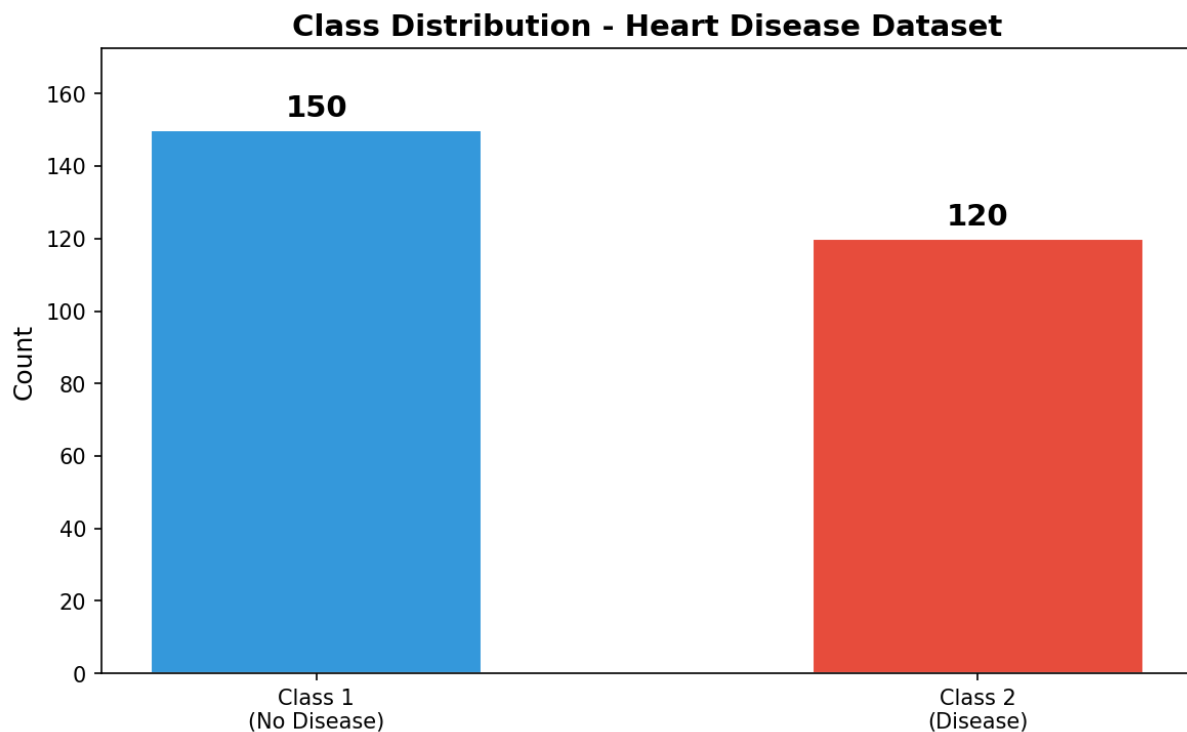
Rank	Feature 1	Feature 2	Correlation (r)
1	opst	dests	+0.6097
2	age	mhr	-0.4022
3	sex	thal	+0.3910
4	mhr	dests	-0.3868
5	mhr	eia	-0.3807

1.7 Top 5 Features Most Correlated with Target (alp2)

Rank	Feature	Correlation	Interpretation
1	thal	+0.5250	Strong positive predictor
2	nmvcf	+0.4553	Strong positive predictor
3	eia	+0.4193	Moderate positive predictor
4	mhr	-0.4185	Moderate negative predictor
5	opst	+0.4180	Moderate positive predictor

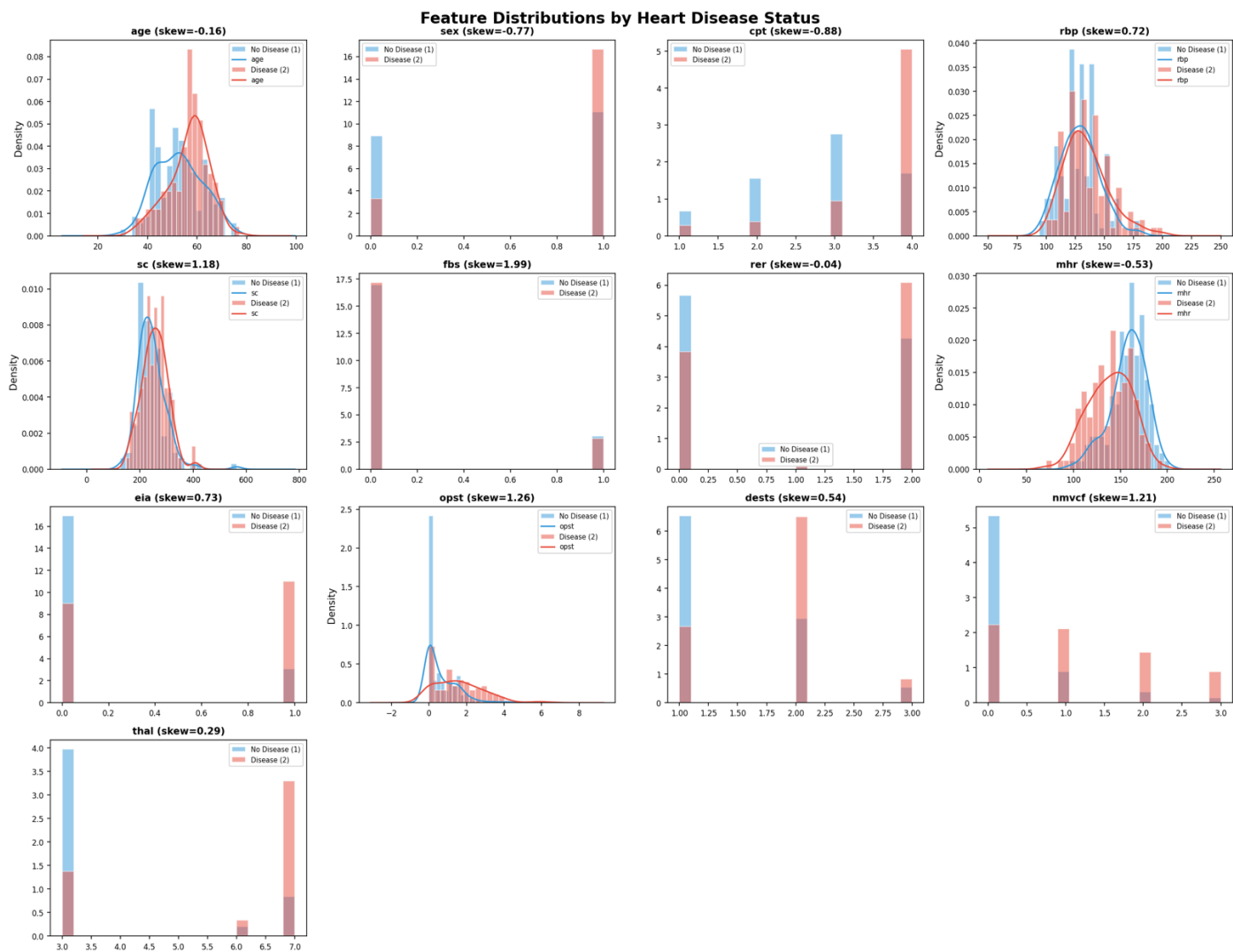
1.8 Class Balance Analysis

Class	Count	Percentage	Status
Class 1 – No Heart Disease	150	55.6%	Majority
Class 2 – Heart Disease	120	44.4%	Minority
Dataset Status	BALANCED	Minority $\geq$ 40%	No resampling needed

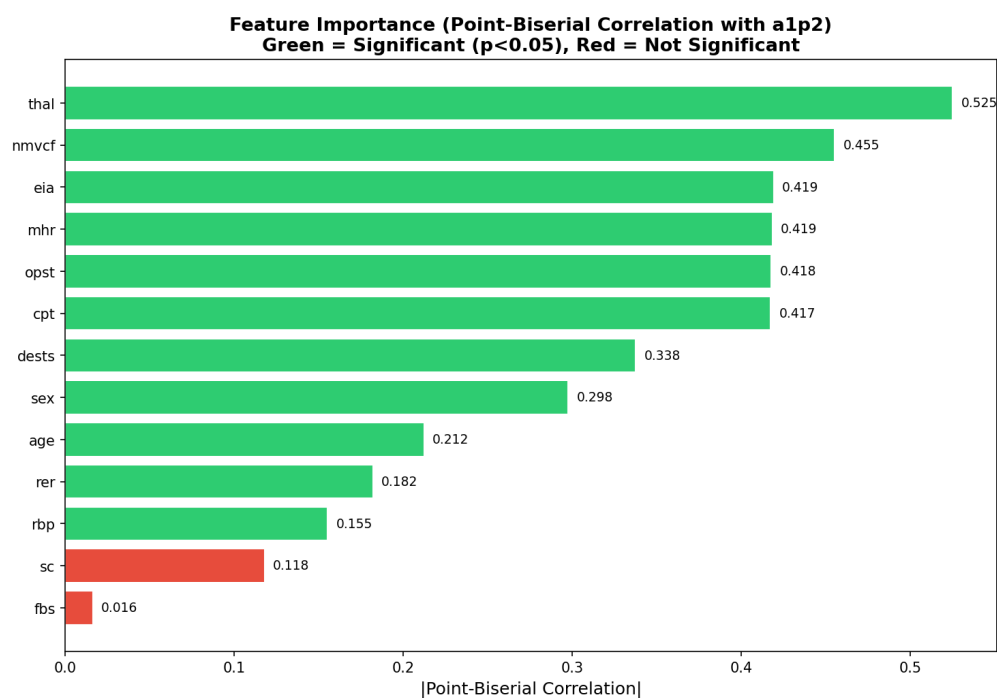


1.9 Distribution Analysis – Skewness and Kurtosis

Feature	Skewness	Kurtosis	Distribution
age	-0.1636	-0.5448	Approx. Normal
sex	-0.7651	-1.4253	Left-Skewed
cpt	-0.8788	-0.2965	Left-Skewed
rbp	0.7226	0.9231	Right-Skewed
sc	1.1837	4.8956	Right-Skewed
fbs	1.9920	1.9826	Right-Skewed
rer	-0.0447	-2.0054	Approx. Normal
mhr	-0.5277	-0.1031	Left-Skewed
eia	0.7289	-1.4797	Right-Skewed
opst	1.2629	1.7593	Right-Skewed
dests	0.5432	-0.6066	Right-Skewed
nmvcf	1.2099	0.2982	Right-Skewed
thal	0.2873	-1.9006	Approx. Normal



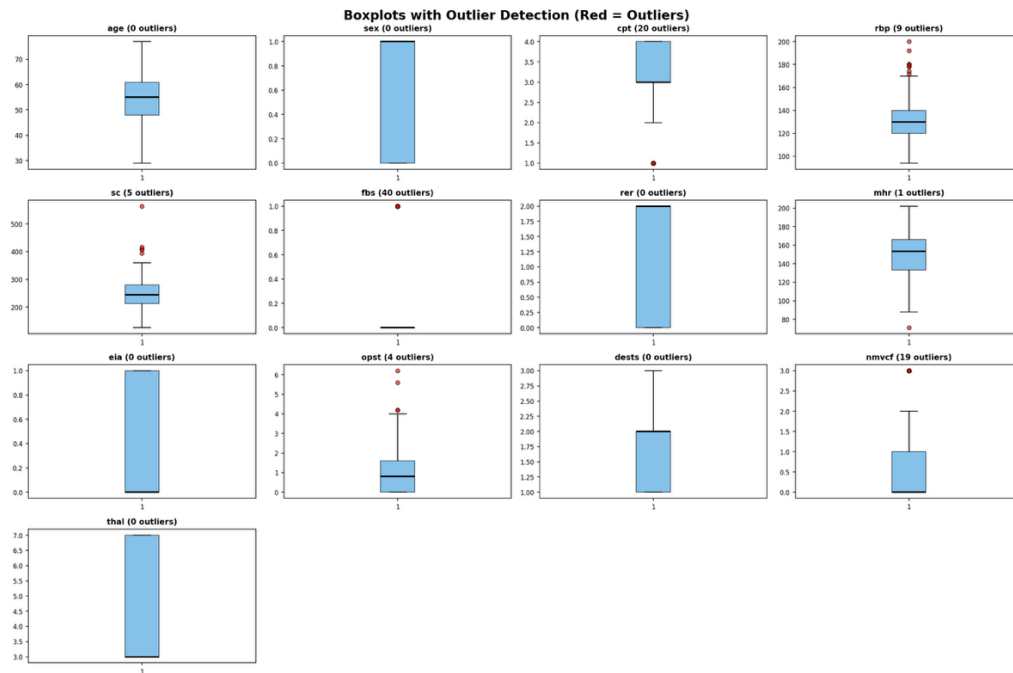
1.10 Feature Importance – Point-Biserial Correlation



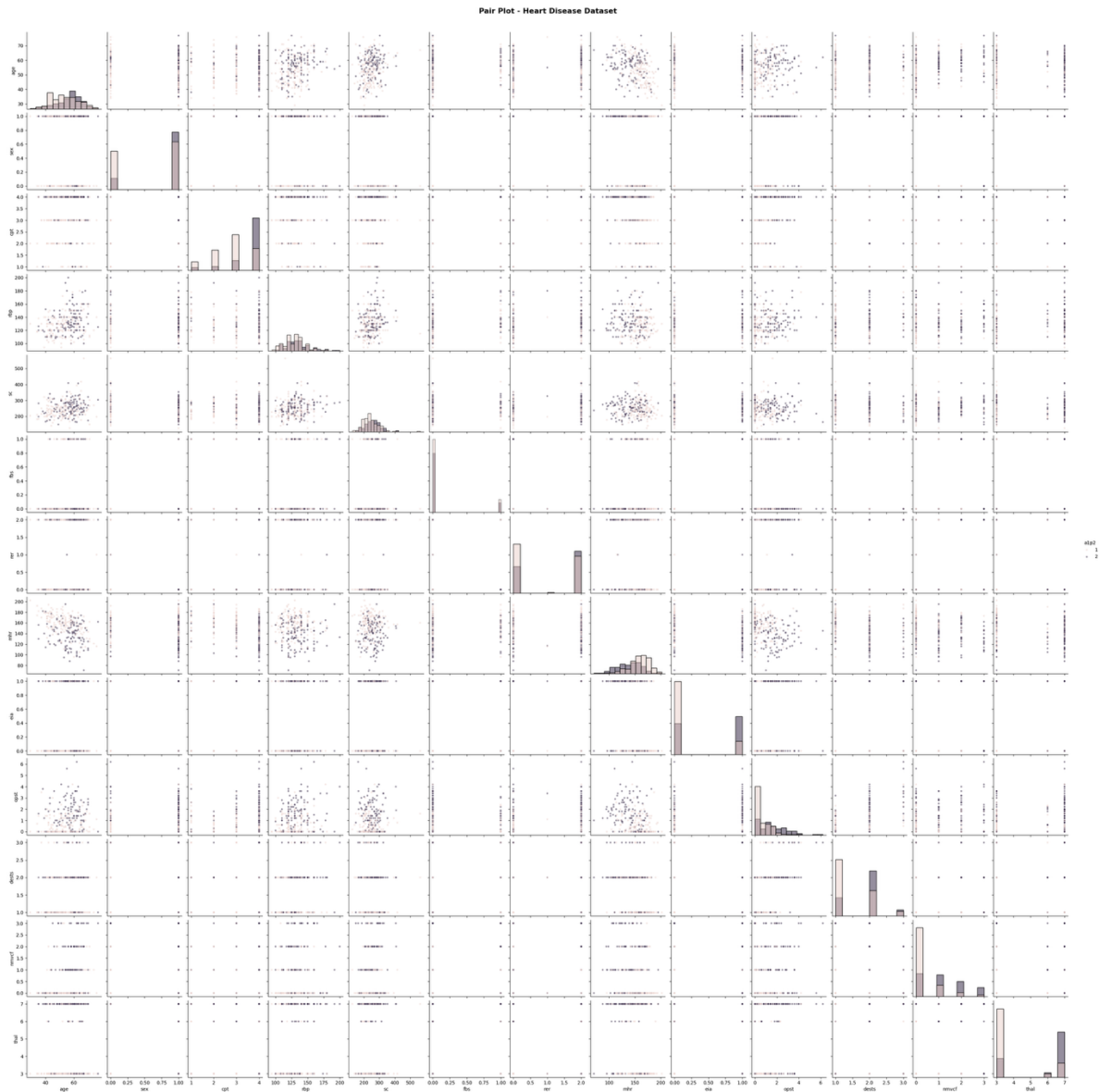
1.11 Outlier Detection (IQR Method)

Feature	# Outliers	% Outliers
age	0	0.0%
sex	0	0.0%
cpt	20	7.4%
rbp	9	3.3%
sc	5	1.9%
fbs	40	14.8%
rer	0	0.0%
mhr	1	0.4%
eia	0	0.0%
opst	4	1.5%
dests	0	0.0%
nmvcf	19	7.0%
thal	0	0.0%

Features with outliers: 7 out of 13



1.12 Pair Plot



1.13 Management Summary - Problem 1

After analyzing the heart disease dataset with 270 patient records and 13 features, here is what I found. The features most strongly linked to heart disease were thal ($r = 0.525$), mvcf ($r = 0.455$), eia ($r = 0.419$), mhr ($r = -0.419$), and opst ($r = 0.418$). Thal being the strongest predictor makes sense since it directly measures heart function through the thallium stress test. I also noticed that opst and mvcf are highly correlated with each other ($r = 0.610$), and age and mhr are negatively correlated ($r = -0.402$), meaning older patients tend to have lower maximum heart rates. The dataset is fairly balanced with 150 patients without heart disease and 120 with it, so we should not need any special balancing techniques. Out of 13 features, 11 turned out to be statistically significant predictors based on p-value testing. Only sc and fbs were not significant, though I kept all features as required. Based on these findings I think machine learning models should be able to predict heart disease pretty well, probably in the range of 0.80 to 0.94 accuracy, with methods like SVM and Random Forest likely doing the best.

PROBLEM 2: MACHINE LEARNING CLASSIFIERS

Problem 2: Machine Learning Classifiers

2.1 Methodology

The heart disease dataset was split into training (80%) and test (20%) sets using `train_test_split` with `random_state=42`, yielding 216 training samples and 54 test samples. All 13 features were retained without elimination or dimensionality reduction (no PCA used). Features were standardized using `StandardScaler` fitted on training data only and applied to both training and test sets to prevent data leakage. Six supervised learning classifiers were trained and evaluated. Combined Accuracy refers to accuracy computed on the full dataset (all 270 samples) using the trained model.

2.2 Classifier Parameters

Classifier	Parameters Used
Perceptron	<code>max_iter=1000, random_state=42, tol=1e-3</code>
Logistic Regression	<code>max_iter=1000, random_state=42, solver=lbfgs</code>
Support Vector Machines	<code>kernel=rbf, C=1.0, gamma=scale, random_state=42</code>
Decision Tree	<code>max_depth=5, random_state=42</code>
Random Forest	<code>n_estimators=100, max_depth=5, random_state=42</code>
K-Nearest Neighbor	<code>n_neighbors=11, metric=euclidean</code>

2.3 Classifier Results

These are all the results I got:

Perceptron Results:

Accuracy: 0.85

Combined Accuracy: 0.84

Logistic Regression Results:

Accuracy: 0.91

Combined Accuracy: 0.86

Support Vector Machines Results:

Accuracy: 0.94

Combined Accuracy: 0.92

Decision Tree Results:

Accuracy: 0.78

Combined Accuracy: 0.90

Random Forest Results:

Accuracy: 0.93

Combined Accuracy: 0.94

K-Nearest Neighbor Results:

Accuracy: 0.91

Combined Accuracy: 0.84

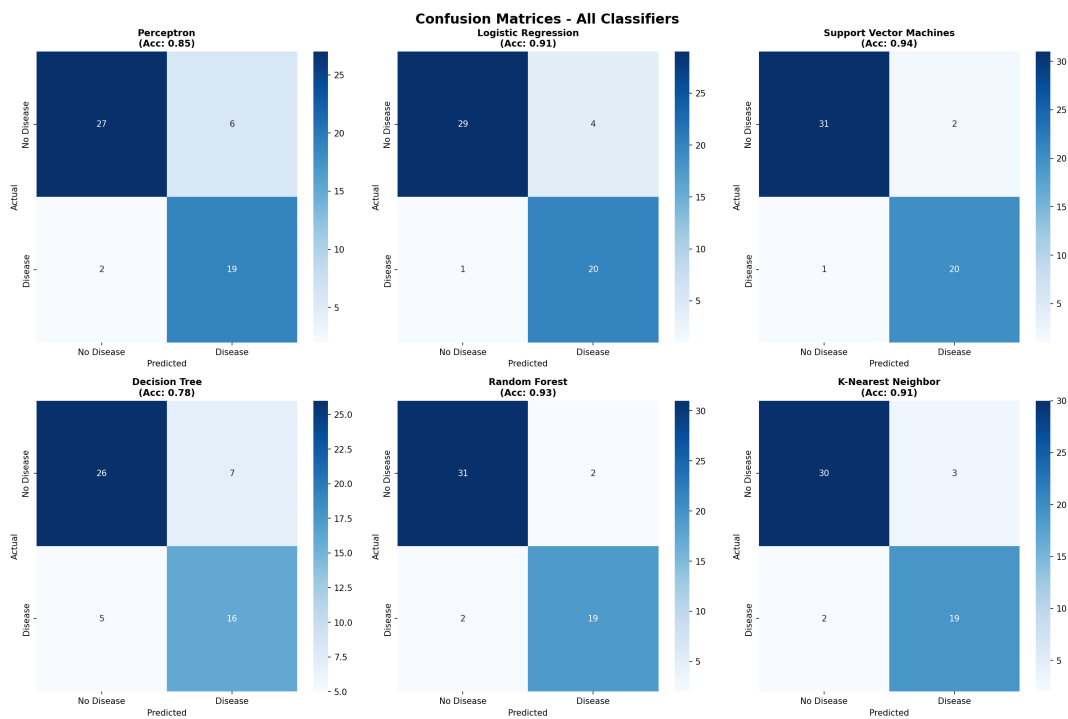
2.4 Comparison Table

Best Method: Support Vector Machines (Test Accuracy: 0.94)

ML Classifier Performance Comparison

Classifier	Test Acc	Combined Acc	CV Mean	CV Std
Perceptron	0.85	0.84	0.7889	+/-0.0764
Logistic Regression	0.91	0.86	0.8000	+/-0.0840
Support Vector Machines	0.94	0.92	0.7741	+/-0.0997
Decision Tree	0.78	0.90	0.6815	+/-0.1219
Random Forest	0.93	0.94	0.7667	+/-0.0833
K-Nearest Neighbor	0.91	0.84	0.8444	+/-0.0569

2.5 Confusion Matrices

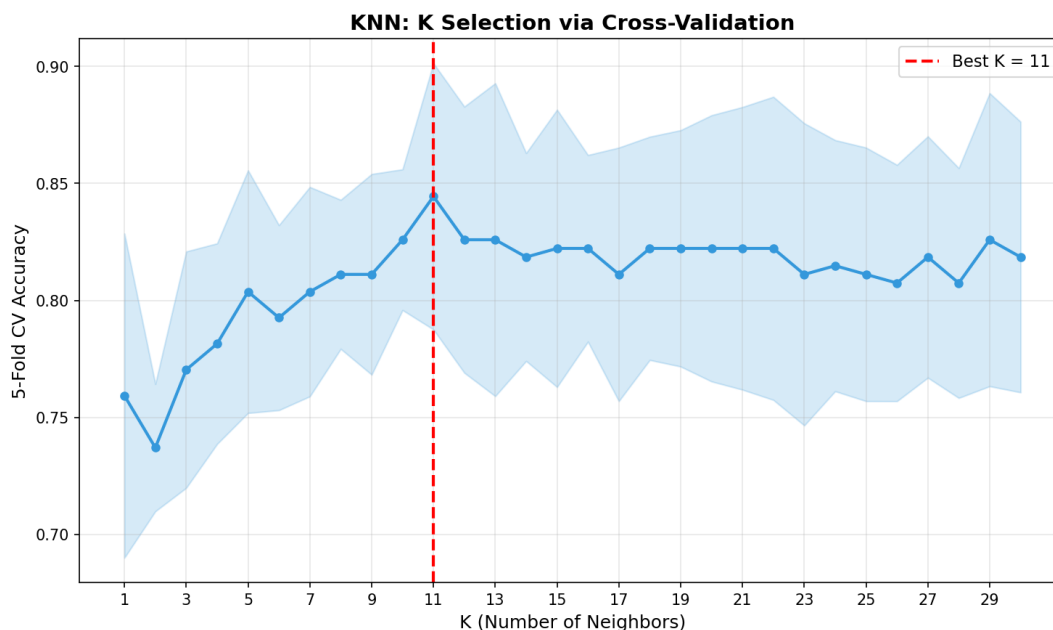


Classifier	TN (No Disease)	FP	FN	TP (Disease)	Accuracy	Precision (Disease)	Recall (Disease)	F1 (Disease)
Perceptron	27	6	2	19	0.85	0.76	0.90	0.83
Logistic Regression	29	4	1	20	0.91	0.83	0.95	0.89
Support Vector Machines	31	2	1	20	0.94	0.91	0.95	0.93
Decision Tree	26	7	5	16	0.78	0.70	0.76	0.73
Random Forest	31	2	2	19	0.93	0.90	0.90	0.90
K-Nearest Neighbor	30	3	2	19	0.91	0.86	0.90	0.88

2.6 KNN K-Selection via Cross-Validation

K values from 1 to 30 were evaluated using 5-fold cross-validation. Optimal K = 11 with CV accuracy of 0.8444.

K	CV Accuracy	K	CV Accuracy	K	CV Accuracy
1	0.7593	11	0.8444 ← BEST	21	0.8222
2	0.7370	12	0.8259	22	0.8222
3	0.7704	13	0.8259	23	0.8111
4	0.7815	14	0.8185	24	0.8148
5	0.8037	15	0.8222	25	0.8111
6	0.7926	16	0.8222	26	0.8074
7	0.8037	17	0.8111	27	0.8185
8	0.8111	18	0.8222	28	0.8074
9	0.8111	19	0.8222	29	0.8259
10	0.8259	20	0.8222	30	0.8185

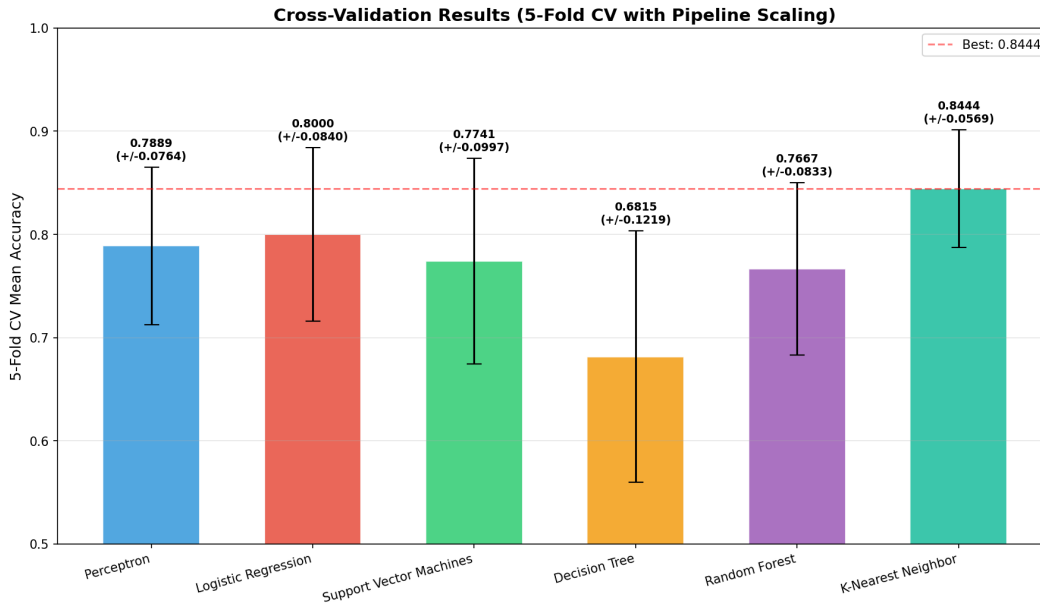


2.7 Cross-Validation Analysis (5-Fold CV)

Classifier	CV Mean Accuracy	CV Std Dev
Perceptron	0.7889	+/- 0.0764
Logistic Regression	0.8000	+/- 0.0840
Support Vector Machines	0.7741	+/- 0.0997
Decision Tree	0.6815	+/- 0.1219
Random Forest	0.7667	+/- 0.0833
K-Nearest Neighbor	0.8444	+/- 0.0569

Best by CV: K-Nearest Neighbor (Mean: 0.8444, Std: 0.0569)

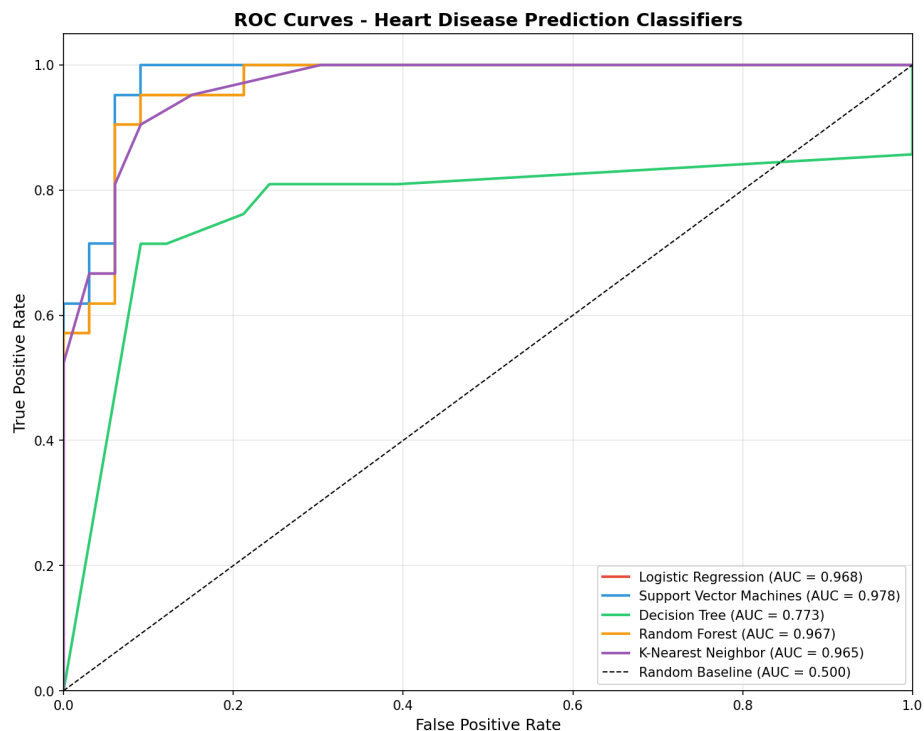
CV scores are lower than single-split test scores because cross-validation provides a more conservative and realistic estimate of generalization performance. The lower standard deviation for KNN (0.0569) also indicates it is the most consistent performer across different data splits.



2.8 ROC Curve Analysis

Classifier	AUC Score
Support Vector Machines	0.9784
Logistic Regression	0.9683
Random Forest	0.9668
K-Nearest Neighbor	0.9654
Decision Tree	0.7727
Perceptron	N/A (no probability support)

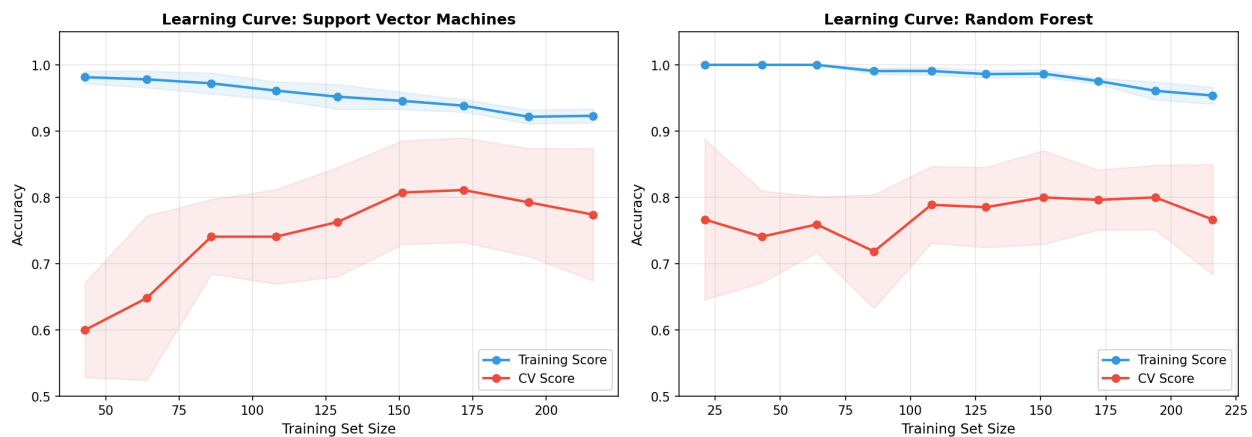
SVM achieved the highest AUC of 0.9784, meaning it has the best ability to discriminate between patients with and without heart disease across all classification thresholds. Decision Tree had the lowest AUC of 0.7727, consistent with its lower test accuracy of 0.78.



2.9 Learning Curve Analysis (Top 2 Models)

Model	Train Score	CV Score	Gap	Assessment
Support Vector Machines	0.9231	0.7741	0.149	Overfitting
Random Forest	0.9537	0.7667	0.187	Overfitting

Both top models show a training-CV gap greater than 0.10, indicating moderate overfitting. This suggests that additional training data could improve generalization performance. The gap is more pronounced for Random Forest (0.187), indicating it memorizes training patterns more than SVM.



2.10 Extended Performance Comparison

Method	Test Acc	Comb Acc	CV Mean	CV Std	AUC
Perceptron	0.85	0.84	0.7889	0.0764	N/A
Logistic Regression	0.91	0.86	0.8000	0.0840	0.9683
Support Vector Machines	0.94	0.92	0.7741	0.0997	0.9784
Decision Tree	0.78	0.90	0.6815	0.1219	0.7727
Random Forest	0.93	0.94	0.7667	0.0833	0.9668
K-Nearest Neighbor	0.91	0.84	0.8444	0.0569	0.9654

2.11 Management Summary – Problem 2

After testing six different machine learning methods on the heart disease dataset, the Support Vector Machine performed the best with a test accuracy of 0.94 and a combined accuracy of 0.92. Random Forest came in second at 0.93 test accuracy. Decision Tree had the lowest accuracy at 0.78, which I think is because it struggles more with the complex relationships in this data compared to ensemble methods. I also ran 5-fold cross-validation to get a more reliable accuracy estimate. Interestingly, KNN with K=11 had the best cross-validation score of 0.844 with a low standard deviation of 0.057, meaning it performed the most consistently across different splits of the data. I found K=11 by testing all values from 1 to 30 and picking the one with the highest CV score. The ROC analysis showed SVM had the highest AUC of 0.978, which means it does the best job of separating disease from non-disease patients. Both SVM and Random Forest showed some overfitting based on the learning curves, with gaps of 0.149 and 0.187 between training and CV scores, so more training data would probably help. Overall I would recommend SVM for AMAPE's prediction system based on its highest test accuracy and AUC, but KNN is also a strong option if consistency across different patient groups is more important.